

ULTRALITE S2 QUICK

One-component, high-performance, highly-deformable, lightweight, rapid-setting and hydrating cementitious adhesive with extended open time, good trowelability, high wetting capacity and extremely high yield, for ceramic tiles and stone, ideal for installing thin porcelain tiles



CLASSIFICATION IN COMPLIANCE WITH ISO 13007-1 AND EN 12004

Ultralite S2 Quick is a cementitious (C), improved (2), rapid-setting (F), with extended open time (E), highly deformable (S2) C2FES2 class adhesive.

WHERE TO USE

- Bonding all types and sizes of ceramic tiles (double-fired, single-fired, porcelain tile, klinker, terracotta, etc.) on internal and external substrates
- Bonding stone on internal and external substrates
- Bonding all types and sizes of thin porcelain tiles on floors and walls, including external façades
- Bonding all types of mosaic on internal and external substrates even in swimming pools

Especially suited for use in large floor areas and for the installation of large format tiles without the need to back butter. Because of its high wetting capacity on the backs of tiles, it is particularly suitable for laying any type of thin porcelain tile (using the double buttering technique), including on thermal insulation systems such as **Mapetherm Tile System**.

SOME APPLICATION EXAMPLES

- Single-fired, double-fired, klinker, terracotta, porcelain and thin porcelain tiles and ceramic and glass mosaic
- Natural stone (marble, granite, etc.) which requires the use of rapid adhesives on;
 - no-slump consistency cementitious screeds, self-levelling cementitious screeds and anhydrite screeds (after applying a suitable primer)
 - heated screeds
 - cementitious render or lime-mortar render
 - gypsum-based render (after applying a suitable primer)
 - plasterboard, precast panels and cement-fibre panels
 - waterproofing membranes made using **Mapelastic**, **Mapelastic Smart**, **Mapelastic AquaDefense**, **Monolastic** and **Mapegum WPS**.
- Rapid restoration work or where surfaces need to be put back into service immediately (areas for public use, motorway service stations, supermarkets, airports, pedestrian areas). Unlike other rapid-setting cementitious adhesives, the extended workability time of **Ultralite S2 Quick** facilitates work carried out in the summer and at higher temperatures
- Overlaying existing ceramic or stone flooring with new ceramic or stone tiles where normal setting adhesives would delay bringing the floor into service due to long setting times
- Installation on marine plywood, wooden agglomerates and old, stable wooden floors
- Installing ceramic and stone on façades, balconies, terraces and paving slabs exposed to direct sunlight and thermal gradients.

- Installation on precast concrete walls and concrete substrates
- Installing thin porcelain tiles on the thermal insulation system **Mapetherm Tile System** (refer to the specific technical notebook)

TECHNICAL CHARACTERISTICS

Ultralite S2 Quick is a grey or white powder made from special binders, selected graded sand and a high quantity of synthetic resin, with added micro-spheres of recycled silica material which help to make the mix lighter, according to a special formulation developed in MAPEI's Research Laboratories, which contributes to the development of eco-sustainable building industry.

The special technology used to manufacture **Ultralite S2 Quick** gives it a low density, a characteristic which offers two main advantages:

1. Bags of **Ultralite S2 Quick** have the same volume but weigh less (15 kg) than bags of conventional cementitious adhesive (25 kg). This ensures easier handling and savings in transport costs
2. Higher yield: Yield is approximately 80% higher than conventional MAPEI cementitious adhesives of class S2.

Ultralite S2 Quick mix has a low viscosity, which makes it easier and quicker to apply. The product's excellent wetting capacity on the back of tiles means that the double buttering technique may be avoided when laying large tiles in internal environments. Its excellent back-face buttering capacity also makes this adhesive particularly suitable for laying thin porcelain tiles. For further details please refer to the relevant technical notebook ("Systems for installing thin porcelain tiles").

The application of **Ultralite S2 Quick** using the double-buttering technique on flat substrates ensures that there are absolutely no gaps in the adhesive on the back face of the slabs, thus avoiding the risk of fracture when in service.

- Excellent capacity of absorbing deformation in the substrate and in the tiles
- Excellent wetting capacity on the back of the tiles
- Adheres perfectly to most materials normally used in the building industry
- **Ultralite S2 Quick** develops a high degree of adhesion in just 2-3 hours from the application at +23°C (8 hours at +10°C), which means floors and coatings may be put into service extremely quickly
- The particularly long open time of the mix compared with other rapid-setting adhesives makes fixing operations more simple, even in the summer and at higher temperatures
- **Ultralite S2 Quick** is in the S2 deformability class according to EN 12004 and ISO 13007-1, and unlike adhesives with the same characteristics, there is no need to add latex, making it simpler and quicker to use

RECOMMENDATIONS

Do not use **Ultralite S2 Quick** in the following cases:

- on metal, rubber, PVC or linoleum surfaces;
- for slabs of marble and natural stone subject to efflorescence or staining;
- for slabs of natural stone highly subject to movement due to humidity, such as green marble, certain types of sandstone and slate (in these cases use **Keralastic**).

Do not add water to the mix once it starts to set.

APPLICATION PROCEDURE

Preparation of the substrate

Substrates must be mechanically strong, free of loose parts, grease, oil, paintwork, wax, etc., and must be sufficiently dry. Cementitious substrates must not shrink after tiles installation. Therefore, in good weather, render must be cured for at least one week per cm of thickness, and cementitious screeds must be cured for at least 28 days, unless they are made using special MAPEI binders for screeds, such as **Mapecem** and **Topcem**, or pre-blended mortars, such as **Mapecem Pronto** and **Topcem Pronto**. If the surface is too hot due to direct sunlight, cool it down with water. Gypsum substrates and anhydrite screeds must be perfectly dry, hard enough for the final intended use and free of dust. They must also be treated with **Primer G** or **Eco Prim T**, while areas subject to high humidity must be primed with **Primer S**. Substrates on which thin porcelain tiles are to be laid must be perfectly flat. Therefore, where necessary, even out the substrate before laying the floor with a self-levelling skimming compound from the MAPEI range.

Preparation of the mix

Blend **Ultralite S2 Quick** with clean water to obtain a smooth, lump-free mix. Let the mix stand for approximately 5 minutes, then blend again. The amount of water required is approximately 5-5.4 litres per 15 kg bag. When blended as described above, the mix remains workable for approximately 50 minutes.

Spreading the mix

Apply **Ultralite S2 Quick** on the substrate using a notched trowel. Use a trowel with a notch size which guarantees complete buttering of the back of the tile. To guarantee a good bond, apply a thin layer of **Ultralite S2 Quick** on the substrate using the smooth side of the trowel and then immediately apply another layer of **Ultralite S2 Quick** to the thickness required using a suitable notched trowel, according to the type and size of the tiles, to guarantee that the backs of the tiles are well buttered. When fixing ceramic tiles to walls and floors externally, for tile sizes larger than 900 cm² and

floors subject to heavy loads, complete buttering of the back of the tiles must be guaranteed. When installing thin porcelain tiles, we recommend that the adhesive is also spread on the backs of the tiles (with a suitable notched trowel) to guarantee there are no gaps, to avoid the risk of fracture when in service.

Laying tiles

Tiles do not need to be wet before they are installed. However, if the backs of the tiles are particularly dusty, wash them by dipping them in clean water. When fixing tiles, apply a firm pressure to guarantee good buttering. The open time for **Ultralite S2 Quick** is at least 30 minutes in normal weather and humidity conditions. When conditions are not ideal (direct sunlight, dry wind, high temperatures, etc.), or if the substrate is particularly absorbent, this time may be reduced to only a few minutes. Keep checking the adhesive to make sure skin does not form on the surface and that it is still fresh. If skin forms, spread the adhesive again with the notched trowel. Do not wet the surface of the adhesive if a skin forms. Water does not dissolve the skin but creates a film which impedes bonding. Final adjustment of the tiles must be carried out within 15 minutes of installation. Tiled finishes fixed using **Ultralite S2 Quick** must be protected from water and rain for at least 3-4 hours and from frost and direct sunlight for at least 24 hours.

Grouting and sealing

Joints may be grouted after 2-3 hours on walls and 2-3 hours on floors. Use a MAPEI cementitious or epoxy grout, available in a wide variety of colours. Seal expansion joints using an appropriate MAPEI sealant.

Set to light foot traffic

Floors are set to light foot traffic after approximately 2-3 hours

Ready to use

Surfaces are ready-to-use after approximately 24 hours.

Cleaning

Clean tools and containers with plenty of water while **Ultralite S2 Quick** is still fresh. Clean the surfaces of coatings using a damp cloth before the adhesive hardens.

PACKAGING

Ultralite S2 Quick is supplied in 15 kg paper bags.

CONSUMPTION

0.8 kg/m² per mm of thickness, equivalent to 1.5-2.5 kg/m²

STORAGE

Ultralite S2 Quick may be stored for up to 12 months in its original packaging in a dry place.

SAFETY INSTRUCTIONS FOR PREPARATION AND APPLICATION

Instruction for the safe use of our products can be found on the latest version of the SDS available from our website www.mapei.ae.

TECHNICAL DATA (typical values) Complies to the following standards: – European EN 12004 (C2FES2) – ISO 13007-1 (C2FES2)	
PRODUCT IDENTIFICATION	
Consistency:	powder
Colour:	grey and white
Bulk density (kg/m³):	900
Dry solids content (%):	100
EMICODE:	EC1 Plus - very low emission
APPLICATION DATA (at +23°C and 50% R.H.)	



Mixing ratio:	100 parts of Ultralite S2 Quick with 33-36% parts of water (by weight)
Consistency of mix:	creamy
Density of mix (kg/m ³):	1,100
pH of mix:	more than 12
Pot life of mix:	50 minutes
Application temperature	from +5°C to +40°C
Open time (according to EN 1346)	≥ 30 minutes
Adjustment time	15 minutes
Grouting tile joints on walls:	after 2-3 hours
Grouting tile joints on floors:	after 2-3 hours
Set to light foot traffic	2-3 hours
Ready-to-use:	24 hours
FINAL PERFORMANCE	
Adhesion according to EN 1348 (N/mm ²): – initial adhesion (after 28 days): – adhesion after application of heat source: – adhesion after immersion in water: – adhesion after freeze-thaw cycles: – initial adhesion after 6 hours:	2.5 2.0 1.2 1.3 0.8
Resistance to alkalis:	Excellent
Resistance to oils:	excellent (poor vegetable oils)
Resistance to solvents:	Excellent
In-service temperature range:	from -30°C to +90°C
Deformability according to EN 12004 and ISO 13007-1:	S2 - highly deformable (> 5 mm)

IMPORTANT NOTES

Whilst we try to ensure that any advice, recommendations or information given in our literature is accurate and correct, we have no control over the circumstances in which our product is used. It is therefore important that installers satisfy themselves that the product and conditions are suitable for the envisaged application. No warranty can be given or responsibility accepted other than, that the product supplied by us will meet our written specification. The installer should ensure that our latest product data and safety information sheets have been consulted prior to use.

Please refer to the current version of the Technical Data Sheet, available from our website www.mapei.com.

LEGAL NOTICE

The contents of this Technical Data Sheet ("TDS") may be copied into other project-related documents, but the resulting document shall not supplement or replace requirements per the TDS in force at the time of the MAPEI product installation. The most up-to-date TDS can be downloaded from our website www.mapei.com.

ANY ALTERATIONS TO THE WORDING OR REQUIREMENTS CONTAINED OR DERIVED FROM THIS TDS EXCLUDES THE RESPONSIBILITY OF MAPEI.

All relevant references for the product are available upon request and from www.mapei.com.

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